

ABHI JAIN

Boston, MA

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EDUCATION

Boston University

September 2022 - May 2027 (expected)

Ph.D. Biostatistics

- Relevant coursework: Estimation Theory, Hypothesis Testing, Linear Models, Correlated Data Analysis, Nonparametric and Semiparametric Data Modeling, Statistical Analysis of Point Process Data, Causal Inference, Machine Learning
- Dissertation: Bayesian nonparametric methods for challenges in causal inference

Wake Forest University

August 2020 - May 2022

M.S. Statistics

- Relevant Coursework: Generalized Linear Models, Bayesian Statistics, Statistical Learning, Time Series, Mathematical Biology, Causal Inference, Stochastic Processes
- Master's Thesis: Using Zero-inflated Models to Estimate the Effect of Air Pollution on Self-Reported Mental Health

Davidson College

August 2014 - May 2018

B.A. Economics; Minor in Applied Mathematics, *cum laude*

- Relevant coursework: Microeconomic Theory, Macroeconomic Theory, Environmental Economics, Multivariable Calculus, Linear Algebra, Econometrics, Probability, Mathematical Statistics, Mathematical Modeling, Health Economics, Game Theory, Computational Economics

EMPLOYMENT

Graduate Research Assistant, Boston University School of Public Health *June 2022 - April 2026*

- Research in spatial statistics and modeling health outcomes
- Lead manuscript development; conduct literature reviews; perform data cleaning and data analysis tasks

Research Assistant, Wake Forest Baptist Health

February 2021 - May 2022

- Conducted research into how healthcare usage has changed with Medicaid expansion
- Constructed datasets used to analyze hospitalization data; produced tables and figures to effectively communicate trends present in hospitalization data

Data Analyst, NORC

June 2018 - May 2022

- Served as a contractor for the Economics Unit of the Office of the Investor Advocate at the Securities and Exchange Commission.
- Developed econometric models to analyze investor decision-making and issues in financial markets
- Cleaned and performed statistical analysis on survey data and financial datasets

RESEARCH INTERESTS

- **Methodology**

Bayesian statistics, Bayesian nonparametrics, spatial statistics, causal inference, causal machine learning, dynamic treatment regimes

- **Applications**

Well-being and health outcomes, food access, neighborhood effects research, air pollution epidemiology, electronic health/medical records data

TEACHING EXPERIENCE

Teaching Fellow

Department of Biostatistics, Boston University School of Public Health

- Survey in Biostatistical Methods (Summer 2026)

Teaching Assistant

Department of Biostatistics, Boston University School of Public Health

- Bayesian Modeling for Biomedical Research & Public Health (Spring 2026)
- Applied Causal Inference in Health Research, Teaching Assistant (Fall 2024)

Department of Mathematics and Statistics, Wake Forest University

- Elementary Probability and Statistics, Teaching Assistant (Fall 2020)

Department of Economics, Davidson College

- Statistics and Basic Econometrics, Teaching Assistant (Spring 2018)

Tutor

Department of Mathematics and Statistics, Wake Forest University

- Math & Stats Center (Fall 2020)

PUBLICATIONS

Alperen, J., LaValley, M., Ni, P., Tripp, T., Emig, D., Patil, P., Mohammed, S., **Jain A.**, Rickles M., Thompson E., Davis B., Sullivan S., Acker M., Dukes K. (2026) Characterizing U.S. Community Well-Being Indices: A Methodological Synthesis and Measurement Systems Perspective. *Population Health Metrics*

Dukes, K., Ni, P., Alperen, J., Cesare, N., LaValley, M., Tripp, T., Lane, K., Mohammed, S., Patil, P., Emig, D., Winter, M., Davis B., Wang B., **Jain A.**, Acker M., and Rickles M. (2025) A national growth mixture modeling analysis of county-level COVID-19 incidence rate trajectories and health inequities during three successive pandemic waves in 2020. *Scientific Reports*.

Jain A., LaValley M., Dukes K., Lane K., Winter M., Spangler K.R., Cesare N., Wang B., Rickles M., and Mohammed S. (2024) Modeling health and well-being measures using ZIP Code spatial neighborhood patterns. *Scientific Reports*.

WORKING PAPERS

Jain A., Lugo N., Bautista-Hurtado D., Zuniga S., Penugonda A., Jean Philippe N., Labossiere S., Pluta K., Mitchell S., Jack B., Fernandez A., Drainoni M., Salloum R., Austad K. (2026+) A Discrete Choice Experiment to Understand Hospital Discharge Preferences of Populations with Non-English Language Preference. *Under revision*.

Jain A., LaValley M., Dukes K., Mohammed S., (2026+) Modeling bounded well-being indices using Bayesian double generalized beta regression with spatial and temporal borrowing. *Under revision*.

Jain A., Jin W., Mohammed S., Lodi S., (2026+) Bayesian causal forests for estimating heterogeneous effects with joint treatments and interference. *In preparation*.

Jain A., Mohammed S., Lodi S., Jin W., (2026+) Joint BART model with latent factors for causal inference in clustered observational studies. *In preparation*.

Jain A., O'Donnell A., Madenci A., Lodi S., (2026+) Practical issues in generating a bootstrap distribution for causal effects of longitudinal treatment strategies. *In preparation*.

CONFERENCE PRESENTATIONS

Conferences (invited, contributed, poster)

Modeling bounded well-being indices using Bayesian double-generalized beta regression with spatial and temporal borrowing. Topic-contributed presentation at 2026 Eastern North American Region Spring Meeting, 2026 March 18, Indianapolis, USA.

Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Topic-contributed presentation at 2025 Eastern North American Region Spring Meeting, 2025 March 24, New Orleans, USA.

Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Topic-contributed presentation at Joint Statistical Meetings, 2024 August 8, Portland, USA.

Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns. Topic-contributed presentation at Joint Statistical Meetings, 2023 August 8, Toronto, Canada.

Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns: a case study on Massachusetts and Georgia. Poster session presentation at New England Statistics Symposium, 2023 June 6, Boston, MA.

Regression-Based Oversampling for Mutual Fund Owners. Contributed poster presentation at Joint Statistics Meetings, 2020 August 4, Virtual.

General audience

Performance of “bag of little bootstrap” in simulated complex longitudinal data. Kolokotronis Circle presentation at CAUSALab, Harvard T.H. Chan School of Public Health, 2025 September 19, Boston, MA.

Modeling well-being using ZIP Code spatial neighborhoods under a Bayesian Beta regression framework. Poster presentation at Boston University Population Health Data Science Poster Session, 2025 September 3, Boston, MA.

Modeling health and well-being measures by incorporating zip-code spatial neighborhood patterns. Poster presentation at Boston University Population Health Data Science Poster Session, 2023 September 7, Boston, MA.

SERVICE

Officer - Boston University Student Chapter of the ASA	<i>March 2024-Present</i>
Student representative - Biostatistics Graduate Education Committee	<i>June 2024-Present</i>
Volunteer academic mentor - Boston Partners in Education	<i>February 2024-December 2025</i>

HONORS & AWARDS

- Distinguished Student Paper Award - *ENAR 2026 Spring Meeting*

- Student Travel Award (Fall 2024) - *Boston University School of Public Health*

SKILLS

Programming Languages and Software

R, Stata, Python, SAS, SQL, Excel, Word, PowerPoint, Visio, Final Cut Pro

Languages

Hindi (speaking)